

Synthetic Control

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Applied Econometrics

- ▶ Another popular research design is the **Regression Discontinuity Design**.
- ▶ In some sense this is a special case of IV regression. (RDD estimates a LATE).
- ▶ Most of this is taken from the JEL Paper by Lee and Lemieux (2010).

- ▶ We have a **running or forcing variable** x such that

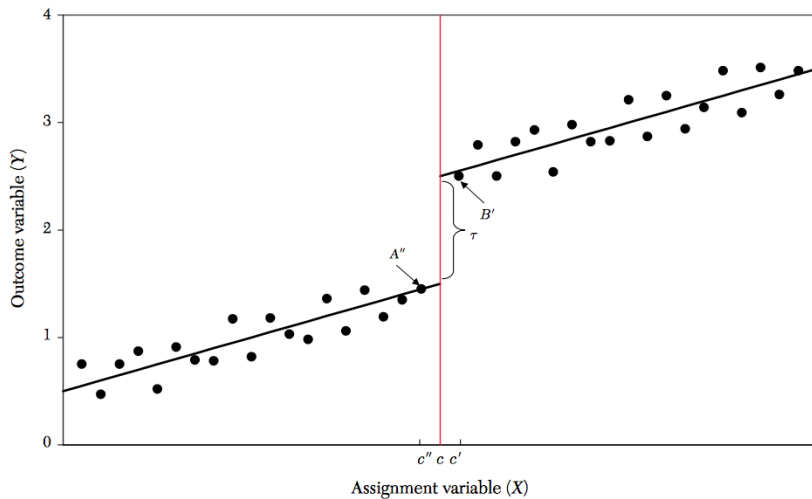
$$\lim_{x \rightarrow c^+} \Pr(D_i | X_i = x) \neq \lim_{x \rightarrow c^-} \Pr(D_i | X_i = x)$$

- ▶ The idea is that there is a **discontinuous jump** in the **probability of being treated**.
- ▶ For now we focus on the **sharp discontinuity**:
 $\Pr(D_i | X_i \geq c) = 1$ and $\Pr(D_i | X_i < c) = 0$
- ▶ There is no single x for which we observe treatment and control.
(Compare to Propensity Score!).

- ▶ Example: a social program is available to people who earned less than \$25,000.
 - If we could compare people earning \$24,999 to people earning \$25,001 we would have as-if random assignment. (MAYBE)
 - But we might not have that many people...
- ▶ We are going to label the **treatment effect** τ_i .

Note: my lack of precision here!
- ▶ The most important assumption is that of **no manipulability** $\tau_i \perp D_i$ in some neighborhood of c .
 - If agents can **choose** x_i we are in trouble: underreporting income, avoiding “possession with intent to distribute” for drugs, etc.

- ▶ The central idea in RDD is that of **continuity**
- ▶ We need that $\mathbb{E}[Y(1) \mid X]$ and $\mathbb{E}[Y(0) \mid X]$ both be continuous at $X = c$.
 - We expect that $Y_i = f(x_i)$ to be a smooth, continuous function of x_i
 - The **only** departure from that is the treatment $\tau_i \cdot I(x_i \geq c)$.
- ▶ We want to be as agnostic as possible about **functional form**
 - Don't want to restrict ourselves to $f(x_i) = \beta_0 + \beta_1 x_i$.
 - The central idea: we know $f(x_i)$ absent the treatment!



RDD uses a set of assumptions distinct from our LATE/IV assumptions. Instead it depends on **continuity**.

- ▶ People just to the left of c are a valid control for those just to the right of c .
- ▶ **This is not a testable assumption** → draw pictures!
- ▶ We could run the regression where $D_i = \mathbf{1}[X_i > c]$.

$$Y_i = \beta_0 + \tau_i \cdot D_i + X_i\beta + \epsilon_i$$

- ▶ This puts a lot of restrictions (linearity) on the relationship between Y and X .
- ▶ Also (without additional assumptions) we only learn about τ_i at the point $X = c$.

First thing to relax is assumption of linearity.

$$Y_i = f(x_i) + \tau \cdot D_i + \epsilon_i$$

This is known as **partially linear model**.

► Three options for $f(x_i)$:

1. Kernels

2. Polynomials: $Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_p x_i^p + \tau D_i + \epsilon_i$.

■ Actually, people suggest different polynomials on each side of cutoff! (Interact everything with D_i).

3. Local Linear/Polynomial Regression (next semester)

► Same objective. Want to flexibly capture what happens on both sides of cutoff.

► Otherwise risk confusing nonlinearity with discontinuity!

RDD: Kernel Boundary Problem

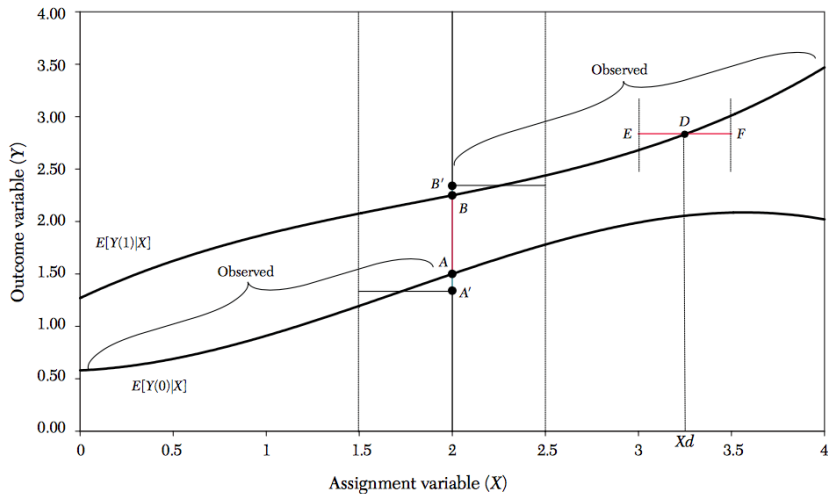


Figure 2. Nonlinear RD

RDD: Polynomial Implementation Details

To make life easier:

- ▶ replace $\tilde{x}_i = x_i - c$.
- ▶ Estimate coefficients $\beta: (1, \tilde{x}, \tilde{x}^2, \dots, \tilde{x}^p)$ and $\tilde{\beta}: (D_i, D_i\tilde{x}, D_i\tilde{x}^2, \dots, D_i\tilde{x}^p)$.
- ▶ Now treatment effect at c just the coefficient on D_i . (We can ignore the interaction terms).
- ▶ If we want treatment effect at $x_i > c$ then we have to account for interactions.
 - Identification away from c is somewhat dubious anyway.
- ▶ Lee and Lemieux (2010) suggest estimating a coefficient on a dummy for each bin in the polynomial regression $\sum_k \phi_k B_k$.
 - Add polynomials until you can satisfy the test that the joint hypothesis test that $\phi_1 = \dots \phi_k = 0$.
 - There are better ways to choose polynomial order...

RDD: Checklist

Most RDD papers follow the same formula (so should yours)

- ▶ Plot of $\Pr[D \mid X]$ so that we can see the discontinuity
- ▶ Plot of $\mathbb{E}[Y \mid X]$ so that we see discontinuity there also
- ▶ Plot of $\mathbb{E}[W \mid X]$ so that we don't see a discontinuity in controls.
- ▶ Density of X (check for manipulation).
- ▶ Show robustness to different “windows”
- ▶ The OLS RDD estimates
- ▶ The Local Linear RDD estimates
- ▶ The polynomial (from each side) RDD estimates
- ▶ An f-test of “bins” showing that the polynomial is flexible enough.

Read Lee and Lemieux (2010) before you get started.

The `rdd` package in R has many of these features.

Looked at incumbency advantage in the US House of Representatives

- ▶ Running variable was vote share in previous election
 - Problem of naive approach: good candidates get lots of votes!
 - Compare outcomes of districts with barely D to barely R .
- ▶ First we plot bin-scatter plots and quartic (from each side) polynomials.
- ▶ Discussion about how to choose bin-scatter bandwidth (CV).

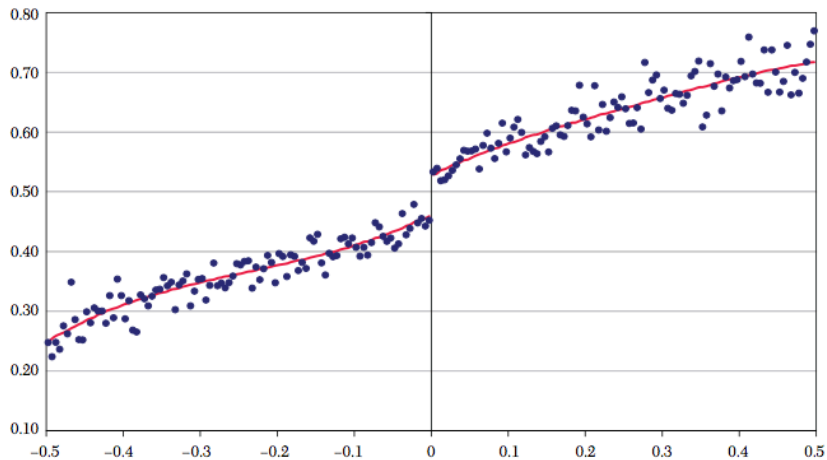


Figure 8. Share of Vote in Next Election, Bandwidth of 0.005 (200 bins)

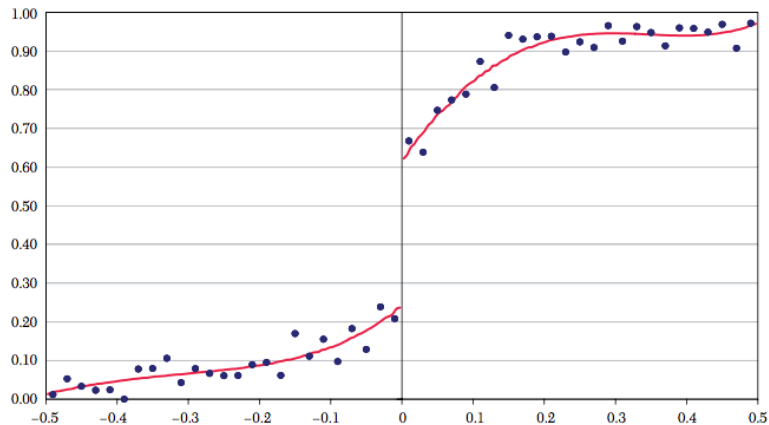


Figure 9. Winning the Next Election, Bandwidth of 0.02 (50 bins)

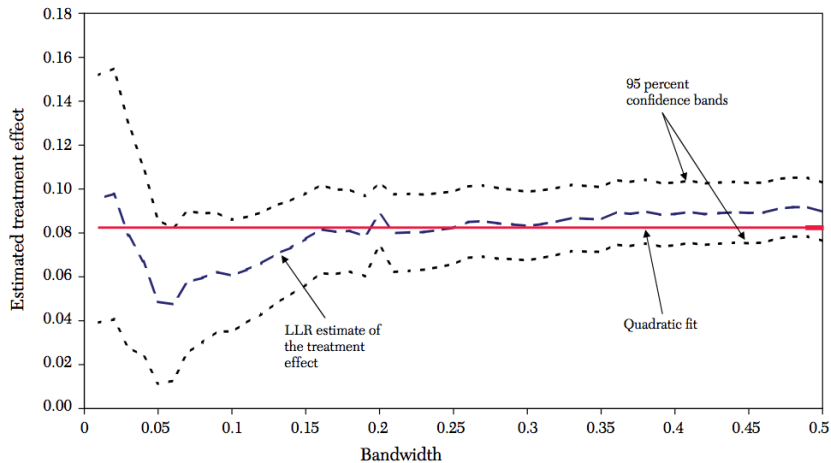


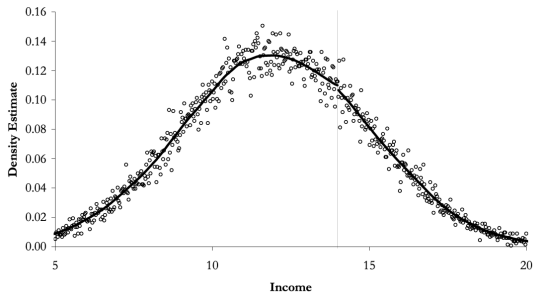
Figure 18. Local Linear Regression with Varying Bandwidth: Share of Vote at Next Election

Luca on Yelp

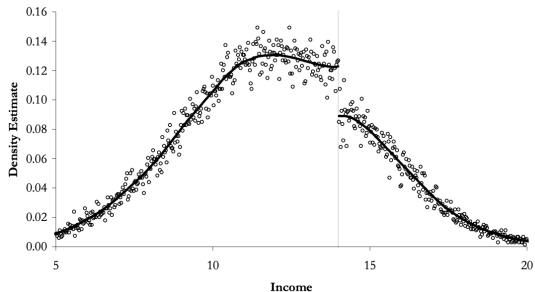
- ▶ Have data on restaurant revenues and yelp ratings.
- ▶ Yelp produces a yelp score (weighted average rating) to two decimals ie: 4.32.
- ▶ Score gets rounded to nearest half star
- ▶ Compare 4.24 to 4.26 to see the impact of an extra half star.
- ▶ Now there are multiple discontinuities: Pool them? Estimate multiple effects?

McCrary (2008) develops a test for manipulation in running variable x

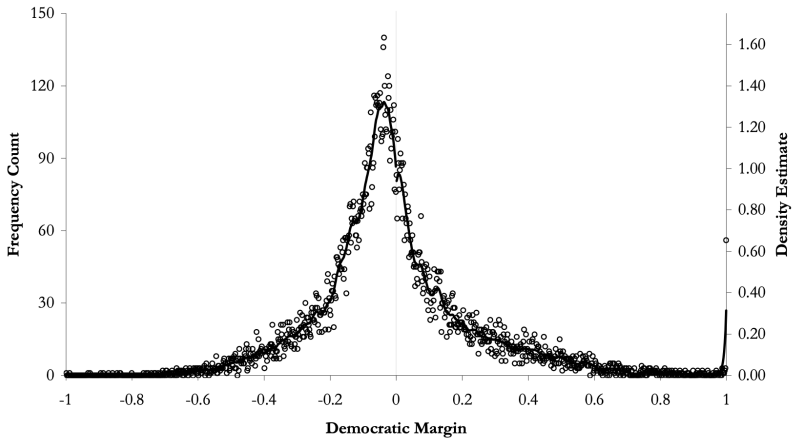
C. Density of Income
with No Pre-Announcement and No Manipulation



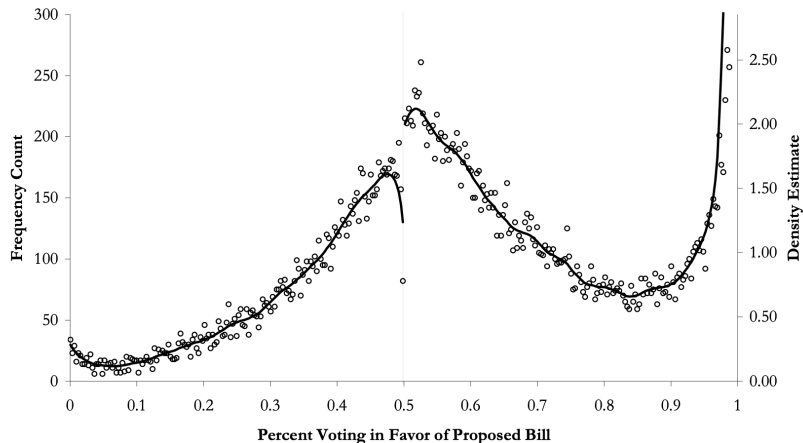
D. Density of Income
with Pre-Announcement and Manipulation



**Figure 4. Democratic Vote Share Relative to Cutoff:
Popular Elections to the House of Representatives, 1900-1990**



**Figure 5. Percent Voting Yeay:
Roll Call Votes, U.S. House of Representatives, 1857-2004**



Extensions

An important extension in the **Fuzzy RD**. Back to where we started:

$$\lim_{x \rightarrow c^+} \Pr(D_i \mid X_i = x) \neq \lim_{x \rightarrow c^-} \Pr(D_i \mid X_i = x)$$

- We need a discontinuous jump in probability of treatment, but it doesn't need to be $0 \rightarrow 1$.

$$\tau_i(c) = \frac{\lim_{x \rightarrow c^+} \mathbb{E}(Y_i \mid X_i = x) - \lim_{x \rightarrow c^-} \mathbb{E}(Y_i \mid X_i = x)}{\lim_{x \rightarrow c^+} \Pr(D_i \mid X_i = x) - \lim_{x \rightarrow c^-} \Pr(d_i \mid X_i = x)} \quad \text{Wald Estimator}$$

- Under sharp RD everyone was a **complier**, now we have some **always takers** and some **never takers** too.
- Now we are estimating the treatment effect only for the population of compliers at $x = c$.
- This should start to look familiar. We are going to do IV!

Related Idea: Kinks

A related idea is that of **kinks**.

- ▶ Instead of a discontinuous jump in the outcome there is a discontinuous jump in β_i on x_i .
- ▶ Often things like tax schedules or government benefits have a kinked pattern.

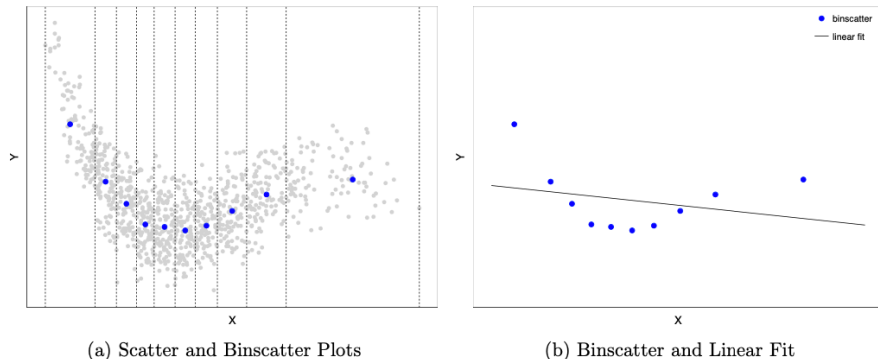
What about binscatter?

All of these RDD plots used something called **binscatter**

- ▶ For the running variable x_i break it up into $J = 20, 50, 100$ bins of equal **density**
- ▶ Use the **quantiles** of x_i .
 - If we have 10 bins, each contains 10% of the sample
 - Like a histogram but with **bins of different width**.
- ▶ Compute the average value: $b_x^j = \mathbb{E}[x_i \mid x_i \in \text{bin}_j]$
- ▶ Compute the average value: $b_y^j = \mathbb{E}[y_i \mid x_i \in \text{bin}_j]$
- ▶ Plot (b_y^j, b_x^j) for each bin j .
- ▶ Often draw a line of best fit through the points

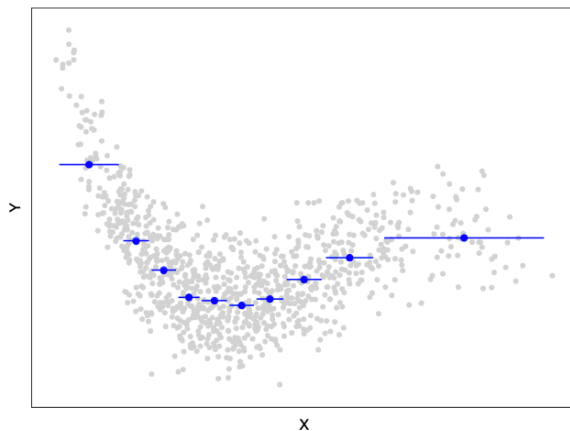
But often we have other covariates!

Figure 1: The basic construction of a binned scatter plot.



Notes. The data is divided into $J = 10$ bins according to the observed x . Within each bin a single dot is plotted at the mean of y for observations falling in the bin. The final plot (b) consists of only these J dots, and the fit from a least squares linear regression of y on x . Constructed using simulated data described in Section SA-5 of the supplemental appendix.

Figure 3: The actual regressogram nonparametric estimator corresponding to a binned scatter plot.



(a) Binned Scatter Plot with Piecewise Constant Fit

Binscatter as Semiparametric Regression:

Cattaneo, Crump, Farrell, Feng (2019)

The right way to think of binscatter is as **semiparametric** or **partially linear** regression

$$y_i = \mu(x_i) + \mathbf{w}_i' \gamma + \epsilon_i, \quad \mathbb{E}[\epsilon_i \mid x_i, \mathbf{w}_i] = 0$$

- ▶ Unless $u(x_i)$ is linear, we can't partial out $y_i - \mathbf{w}_i' \gamma$ via Frisch-Lovell-Waugh
- ▶ Most binscatter software does this wrong. Use binsreg in R.

Thanks
